Subject: ESTIMATION, SPECIFICATIONS, COSTING AND VALUATION

Subject Code: 23A01507

LABORATORY MANUAL

Estimation, Specifications, Costing and Valuation Lab (B.Tech Civil, JNTUA R23)

Course Objectives

- Understand the various methods and types of estimates used in civil engineering projects.
- Develop detailed estimates for single and multi-storey buildings using standard estimation methods.
- Analyse rate analysis, abstract estimation and bill preparation as per standard procedures.
- Prepare detailed specifications and tender documents for construction works.
- Evaluate the valuation, cost escalation and value analysis of buildings.

Course Outcomes

- Apply estimation techniques to prepare detailed estimates for various construction projects.
- Develop abstract estimates and rate analysis for different civil engineering works.
- Analyse the preparation of measurement books and bill preparation as per AP State Government procedures.
- Create detailed specifications and tender documents for construction projects.
- Assess building valuation, cost escalation and value analysis techniques.

Note All the 10 exercises are given here. One residential building plan is used throughout (Exercises 2 to 10) so that all the quantities, rates and amounts are connected. All dimensions, rates and values are sample (assumed) values.

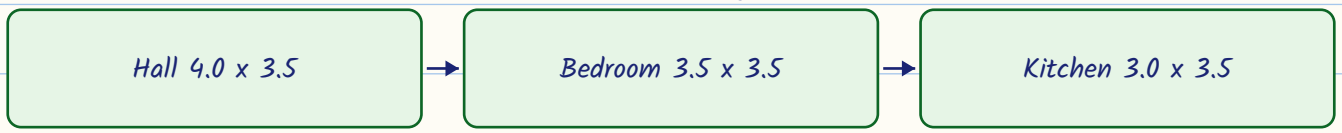
List of Experiments

No	Experiment
1	Activity Based Learning of Methods and Types of Estimates
2	Detailed Estimate of a Single-Storied Residential Building by Wall to Wall Method
3	Detailed Estimate of a Single-Storied Residential Building by Centre Line Method (Earthwork, Foundations, Super Structure, Fittings including Sanitary and Electrical Fittings and Paintings)
4	Detailed Estimate of a Two Storied Residential Building by Centre Line Method
5	Activity Based Learning of Estimate Data and Rate Analysis
6	Abstract Estimate for the Detailed Estimate in Exercise No. 3
7	Abstract Estimate for the Detailed Estimate in Exercise No. 4
8	Writing of Measurement Book and Bill Preparation as per AP State Govt. Procedure
9	Writing Detailed Specifications and Preparing a Model Tender Document
10	Activity Based Learning for Valuation of Buildings, Cost Escalation Procedures and Value Analysis

The Building Used in the Exercises

- Single-storied residential building with three rooms in a row (Hall, Bedroom, Kitchen), flat RCC roof, with a staircase added on the rear side in the two-storied case.
- **Clear room sizes** Hall 4.0 m x 3.5 m, Bedroom 3.5 m x 3.5 m, Kitchen 3.0 m x 3.5 m. All walls are 230 mm thick brick masonry in CM 1:6.

- **Centre line dimensions** length = 4.0 + 3.5 + 3.0 + 3 x 0.23 = 11.19 m and breadth = 3.5 + 0.23 = 3.73 m. Out to out = 11.42 m x 3.96 m.
- **Number of walls** 2 long walls (front and back) and 4 short walls (2 end walls and 2 partition walls).
- **Foundation (single storey)** trench 0.90 m wide and 1.00 m deep, PCC 1:4:8 bed 0.30 m, two brick footing steps (0.69 m and 0.46 m wide, 0.30 m each), basement wall 0.23 m up to plinth level (plinth 0.45 m above ground).
- Height of wall from plinth to roof slab bottom = 3.15 m; RCC slab 120 mm thick.
- **Openings** D1 external door 1.00 x 2.10 (1 No), D2 internal doors 0.90 x 2.10 (2 Nos), W windows 1.50 x 1.20 (3 Nos), V ventilator 0.60 x 0.45 (1 No). RCC lintels 230 x 150 mm with 150 mm bearing on each side.



Experiment 1 – Activity Based Learning of Methods and Types of Estimat

Aim

To learn the different types of estimates and the methods of estimation used for civil engineering works, through classroom activities.

Requirements

- Plan and section of a building, estimate format sheets, calculator, SSR (Standard Schedule of Rates) of the State, IS 1200 (method of measurement of building works).

Theory

An estimate is the calculation of the quantities and probable cost of a work before it is executed. It is needed to know the cost, to get administrative and

technical sanction, to invite tenders and to control expenditure.

Types of estimates

Type	Meaning and use
Preliminary / approximate estimate	Rough cost found early from plinth area, cubic content or unit rates; used for administrative approval (accuracy about 10 to 15 %)
Detailed estimate	Item wise quantities from detailed drawings, with rates; used for technical sanction and tender
Abstract estimate	Summary of quantities with rates and amounts of each item along with contingencies and establishment charges
Revised estimate	Prepared when the sanctioned estimate is likely to be exceeded by more than 5 %
Supplementary estimate	Prepared for additional works found necessary during execution
Annual repair / maintenance estimate	Prepared for yearly repairs and maintenance of existing buildings
Complete estimate	Gives the total cost of a work including land, buildings, services and furniture

Methods of estimation

- **Plinth area method** cost = plinth area x plinth area rate. Quick, used for approximate estimates.
- **Cubic content method** cost = volume (length x breadth x height from foundation top to roof) x cubic rate.
- **Unit base method** cost = number of units x cost per unit (per bed in hospital, per student in school, per seat in hall).
- **Bay method** for buildings with repeated bays (factories, godowns).

- **Detailed method** (a) Long wall and short wall (wall to wall) method and (b) Centre line method.
- Service unit method and Approximate quantity method (cost of services, for example roads per km).

Activity 1 – Approximate estimates for the building of this manual

- Plinth area = out to out length x breadth = 11.42 x 3.96 = 45.22 m2.
- Plinth area rate assumed = Rs. 19,500 per m2. Approximate cost = 45.22 x 19,500 = Rs. 8,81,852.40.
- Cubic content = plinth area x height (foundation top to roof = 4.05 m) = 183.15 m3. Cubic rate assumed Rs. 4,700 per m3. Cost = Rs. 8,60,823.61.
- For comparison the detailed estimate of Exercise 6 gives Rs. 9,56,671.43; the approximate value by plinth area method differs by -7.8 %.

Activity 2 – Identify the type of estimate

Situation	Type of estimate
Cost is found early using a plinth area rate	Preliminary (approximate) estimate
Quantities are worked out item wise from drawings	Detailed estimate
Quantities and rates are tabulated with a total for sanction	Abstract estimate
Sanctioned amount is likely to exceed by 8 %	Revised estimate
A compound wall is added after sanction of the building	Supplementary estimate
Yearly white washing and repair of a school	Annual maintenance estimate

Result

- The types and methods of estimates were studied and the approximate cost of the sample building was found as Rs. 8,81,852.40 (plinth area method).

Experiment 2 - Detailed Estimate by Wall to Wall (Long Wall and Short W

Aim

To prepare the detailed estimate for the single-storied residential building by the long wall and short wall (wall to wall) method.

Requirements

- Plan, section and specifications of the building, IS 1200, estimate sheets, calculator.

Theory

- In this method the walls are divided into long walls (running in the length direction) and short walls (running in the breadth direction).
- Long wall length (out to out) = centre line length + half width of the foundation (or wall) on each side = L(c/c) + w. Short wall length (in to in) = centre line breadth - w.
- Quantity = No x length x breadth x depth (or height).
- The same quantity is obtained by the centre line method if the junction corrections are done correctly.

Procedure

- **Take the centre line dimensions** L = 11.19 m, B = 3.73 m. Number of long walls = 2, short walls = 4.
- For each item find the length of long wall and short wall by using the width of that item (trench width, footing width or wall thickness).
- Enter in the measurement sheet as No x L x B x D, work out the quantity and total it.
- Deduct the openings and lintels from the superstructure brickwork.

Measurement sheet (single storey)

Sl	Particulars of item	No	L (m)	B (m)	D / H (m)	Quantity
1	Earthwork excavation in foundation (m3)					
	Long walls (front and back)	2	12.09	0.90	1.00	21.76
	Short walls (2 end + 2 partition)	4	2.83	0.90	1.00	10.19
	Total quantity					31.95
2	PCC 1:4:8 bed in foundation (m3)					
	Long walls (front and back)	2	12.09	0.90	0.30	6.53
	Short walls (2 end + 2 partition)	4	2.83	0.90	0.30	3.06
	Total quantity					9.59
3	Brickwork in CM 1:6, footing step 1 (width 0.69 m) (m3)					
	Long walls (front and back)	2	11.88	0.69	0.30	4.92
	Short walls (2 end + 2 partition)	4	3.04	0.69	0.30	2.52
	Total quantity					7.44
4	Brickwork in CM 1:6, footing step 2 (width 0.46 m) (m3)					
	Long walls (front and back)	2	11.65	0.46	0.30	3.22
	Short walls (2 end + 2 partition)	4	3.27	0.46	0.30	1.81
	Total quantity					5.02
5	Brickwork in CM 1:6 in basement up to plinth level (m3)					
	Long walls (front and back)	2	11.42	0.23	0.55	2.89
	Short walls (2 end + 2 partition)	4	3.50	0.23	0.55	1.77
	Total quantity					4.66
6	Earth filling in plinth, 0.35 m (m3)					
	Floor area (clear) 36.75 m2 x 0.35 m	1	-	36.75	0.35	12.86

	Total quantity					12.86
7	DPC 25 mm thick, CM 1:3 at plinth level (m2)					
	Long walls	2	11.42	0.23	-	5.25
	Short walls	4	3.50	0.23	-	3.22
	Total quantity					8.47
8	Brickwork in CM 1:6 in superstructure (m3)					
	Ground floor: long walls	2	11.42	0.23	3.15	16.55
	Ground floor: short walls	4	3.50	0.23	3.15	10.14
	Deduct D1 (external door)	1	1.00	0.23	2.10	-0.48
	Deduct D2 (internal door)	2	0.90	0.23	2.10	-0.87
	Deduct W (window)	3	1.50	0.23	1.20	-1.24
	Deduct V (ventilator)	1	0.60	0.23	0.45	-0.06
	Deduct RCC lintels	1	10.00	0.23	0.15	-0.35
	Total quantity					23.69
9	RCC 1:2:4 in lintels (m3)					
	Lintels, 1 floor(s)	1	10.00	0.23	0.15	0.35
	Total quantity					0.35
10	RCC 1:2:4 in roof / floor slab, 120 mm thick (m3)					
	Slab 11.42 x 3.96 x 0.12, 1 Nos	1	11.42	3.96	0.12	5.43
	Total quantity					5.43
11	Reinforcement steel Fe 500 @ 100 kg per m3 of RCC (kg)					
	RCC 5.772 m3 x 100 kg	1	-	-	-	577.2
	Total quantity					577.18

12	Centering and shuttering (m2)					
	Slab soffit 36.75 m2, 1 Nos	1	-	36.75	-	36.75
	Slab edges 30.76 x 0.12, 1 Nos	1	30.76	0.12	-	3.69
	Lintels 10.0 x (0.23 + 2 x 0.15), 1 floor(s)	1	10.00	0.53	-	5.30
	Total quantity					45.74
13	Cement concrete flooring 40 mm thick (1:2:4) (m2)					
	Floor area, 1 floor(s)	1	-	36.75	-	36.75
	Total quantity					36.75
14	Plastering 12 mm thick, CM 1:6, internal walls (m2)					
	GF Hall (4.0 x 3.5), perimeter 15.00	1	15.00	-	3.15	47.25
	GF Bedroom (3.5 x 3.5), perimeter 14.00	1	14.00	-	3.15	44.10
	GF Kitchen (3.0 x 3.5), perimeter 13.00	1	13.00	-	3.15	40.95
	Deduct D1 (external door), 1 No x 1 face(s)	1	1.00	-	2.10	-2.10
	Deduct D2 (internal door), 2 No x 2 face(s)	4	0.90	-	2.10	-7.56
	Deduct W (window), 3 No x 1 face(s)	3	1.50	-	1.20	-5.40
	Deduct V (ventilator), 1 No x 1 face(s)	1	0.60	-	0.45	-0.27
	Total quantity					116.97
15	Plastering 12 mm thick, CM 1:6, ceilings (m2)					
	Ceiling area, 1 floor(s)	1	-	36.75	-	36.75
	Total quantity					36.75
16	Plastering 15 mm thick, CM 1:6, external (m2)					
	External face, perimeter 30.76 x 3.6	1	30.76	-	3.60	110.74

	Deduct openings (D1, D2 x1 face, W, V) x 1	1	-	-	-	-11.55
	Total quantity					99.19
17	Emulsion paint, 2 coats over primer (walls and ceilings) (m2)					
	Internal plaster + ceiling plaster areas	1	-	-	-	153.72
	Total quantity					153.72
18	Weather-proof exterior paint, 2 coats (m2)					
	External plaster area	1	-	-	-	99.19
	Total quantity					99.19
19	Door D1 (panelled, teak, with frame and fixtures) (m2)					
	1 x 1 No x 1.00 x 2.10	1	-	-	-	2.10
	Total quantity					2.10
20	Door D2 (flush, with frame and fixtures) (m2)					
	1 x 2 No x 0.90 x 2.10	1	-	-	-	3.78
	Total quantity					3.78
21	Window W (wooden, glazed with grill) (m2)					
	1 x 3 No x 1.50 x 1.20	1	-	-	-	5.40
	Total quantity					5.40
22	Ventilator V (wooden frame with louvers) (m2)					
	1 x 1 No x 0.60 x 0.45	1	-	-	-	0.27
	Total quantity					0.27
23	Indian / European WC with flush tank (No)					

	Counted from plan, 1 No	1	-	-	-	1
	Total quantity					1
24	Wash basin with pedestal and tap (No)					
	Counted from plan, 1 No	1	-	-	-	1
	Total quantity					1
25	Kitchen sink (stainless steel, with drainboard) (No)					
	Counted from plan, 1 No	1	-	-	-	1
	Total quantity					1
26	CP tap / bib cock (15 mm) (No)					
	Counted from plan, 4 No	4	-	-	-	4
	Total quantity					4
27	Water supply and drainage pipes (lump sum, per floor) (LS)					
	Counted from plan, 1 LS	1	-	-	-	1
	Total quantity					1
28	Light point (concealed wiring) (point)					
	Counted from plan, 9 point	9	-	-	-	9
	Total quantity					9
29	Fan point (concealed wiring) (point)					
	Counted from plan, 3 point	3	-	-	-	3
	Total quantity					3
30	Power socket 5 A (with wiring) (point)					
	Counted from plan, 6 point	6	-	-	-	6
	Total quantity					6
31	Power socket 15 A (with wiring) (point)					
	Counted from plan, 1 point	1	-	-	-	1

	Total quantity					1
32	Distribution board with MCBs (No)					
	Counted from plan, 1 No	1	-	-	-	1
	Total quantity					1

Result

- **Quantities of the main items** earthwork 31.95 m3, PCC 9.59 m3, superstructure brickwork 23.69 m3.
- These agree with the centre line method of Exercise 3 (earthwork 31.95, PCC 9.58, brickwork 23.69 m3), which checks the work.

Experiment 3 – Detailed Estimate by Centre Line Method (Single Storey)

Aim

To prepare the detailed estimate for the single-storied residential building by the centre line method, including earthwork, foundations, superstructure, fittings (sanitary and electrical) and paintings.

Requirements

- Same as Exercise 2.

Theory

- Total centre line length = sum of centre lengths of all walls - (number of junctions x half the width of the item under consideration).
- Sum of centre lengths = 2 x (11.19 + 3.73) + 2 x 3.73 = 29.84 + 7.46 = 37.30 m. There are 4 junctions (partition walls meeting the long walls), so the corrected length for an item of width w is L(w) = 37.30 - 4 x (w/2) = 37.30 - 2w.
- **For example the trench (w = 0.90)** L = 37.30 - 1.80 = 35.50 m; for 230 mm wall :: L = 37.30 - 0.46 = 36.84 m.

Formula Quantity = L(w) x width x depth (or height). The length changes for each different width of foundation.

Procedure

- Find the corrected centre line length for each width (trench 0.90, footing 0.69 and 0.46, wall 0.23).
- Work out earthwork, PCC, brickwork in footings and basement, DPC, plinth filling, superstructure, RCC, flooring, plastering, doors and windows, painting and fittings.
- Deduct the door, window, ventilator openings and lintels from the superstructure brickwork.

Measurement sheet (single storey)

Sl	Particulars of item	No	L (m)	B (m)	D / H (m)	Quantity
1	Earthwork excavation in foundation (m3)					
	Earthwork (centre line)	1	35.50	0.90	1.00	31.95
	Total quantity					31.95
2	PCC 1:4:8 bed in foundation (m3)					
	PCC (centre line)	1	35.50	0.90	0.30	9.58
	Total quantity					9.58
3	Brickwork in CM 1:6, footing step 1 (width 0.69 m) (m3)					
	Footing 1 (centre line)	1	35.92	0.69	0.30	7.44
	Total quantity					7.44
4	Brickwork in CM 1:6, footing step 2 (width 0.46 m) (m3)					
	Footing 2 (centre line)	1	36.38	0.46	0.30	5.02
	Total quantity					5.02

5	Brickwork in CM 1:6 in basement up to plinth level (m3)					
	Basement (centre line)	1	36.84	0.23	0.55	4.66
	Total quantity					4.66
6	Earth filling in plinth, 0.35 m (m3)					
	Floor area (clear) 36.75 m2 x 0.35 m	1	-	36.75	0.35	12.86
	Total quantity					12.86
7	DPC 25 mm thick, CM 1:3 at plinth level (m2)					
	DPC (centre line)	1	36.84	0.23	-	8.47
	Total quantity					8.47
8	Brickwork in CM 1:6 in superstructure (m3)					
	Ground floor: gross masonry (centre line)	1	36.84	0.23	3.15	26.69
	Deduct D1 (external door)	1	1.00	0.23	2.10	-0.48
	Deduct D2 (internal door)	2	0.90	0.23	2.10	-0.87
	Deduct W (window)	3	1.50	0.23	1.20	-1.24
	Deduct V (ventilator)	1	0.60	0.23	0.45	-0.06
	Deduct RCC lintels	1	10.00	0.23	0.15	-0.35
	Total quantity					23.69
9	RCC 1:2:4 in lintels (m3)					
	Lintels, 1 floor(s)	1	10.00	0.23	0.15	0.35
	Total quantity					0.35
10	RCC 1:2:4 in roof / floor slab, 120 mm thick (m3)					
	Slab 11.42 x 3.96 x 0.12, 1 Nos	1	11.42	3.96	0.12	5.43
	Total quantity					5.43

11	Reinforcement steel Fe 500 @ 100 kg per m3 of RCC (kg)					
	RCC 5.772 m3 x 100 kg	1	-	-	-	577.2
	Total quantity					577.18
12	Centering and shuttering (m2)					
	Slab soffit 36.75 m2, 1 Nos	1	-	36.75	-	36.75
	Slab edges 30.76 x 0.12, 1 Nos	1	30.76	0.12	-	3.69
	Lintels 10.0 x (0.23 + 2 x 0.15), 1 floor(s)	1	10.00	0.53	-	5.30
	Total quantity					45.74
13	Cement concrete flooring 40 mm thick (1:2:4) (m2)					
	Floor area, 1 floor(s)	1	-	36.75	-	36.75
	Total quantity					36.75
14	Plastering 12 mm thick, CM 1:6, internal walls (m2)					
	GF Hall (4.0 x 3.5), perimeter 15.00	1	15.00	-	3.15	47.25
	GF Bedroom (3.5 x 3.5), perimeter 14.00	1	14.00	-	3.15	44.10
	GF Kitchen (3.0 x 3.5), perimeter 13.00	1	13.00	-	3.15	40.95
	Deduct D1 (external door), 1 No x 1 face(s)	1	1.00	-	2.10	-2.10
	Deduct D2 (internal door), 2 No x 2 face(s)	4	0.90	-	2.10	-7.56
	Deduct W (window), 3 No x 1 face(s)	3	1.50	-	1.20	-5.40
	Deduct V (ventilator), 1 No x 1 face(s)	1	0.60	-	0.45	-0.27
	Total quantity					116.97
15	Plastering 12 mm thick, CM 1:6, ceilings (m2)					
	Ceiling area, 1 floor(s)	1	-	36.75	-	36.75

	Total quantity					36.75
16	Plastering 15 mm thick, CM 1:6, external (m2)					
	External face, perimeter 30.76 x 3.6	1	30.76	-	3.60	110.74
	Deduct openings (D1, D2 x1 face, W, V) x 1	1	-	-	-	-11.55
	Total quantity					99.19
17	Emulsion paint, 2 coats over primer (walls and ceilings) (m2)					
	Internal plaster + ceiling plaster areas	1	-	-	-	153.72
	Total quantity					153.72
18	Weather-proof exterior paint, 2 coats (m2)					
	External plaster area	1	-	-	-	99.19
	Total quantity					99.19
19	Door D1 (panelled, teak, with frame and fixtures) (m2)					
	1 x 1 No x 1.00 x 2.10	1	-	-	-	2.10
	Total quantity					2.10
20	Door D2 (flush, with frame and fixtures) (m2)					
	1 x 2 No x 0.90 x 2.10	1	-	-	-	3.78
	Total quantity					3.78
21	Window W (wooden, glazed with grill) (m2)					
	1 x 3 No x 1.50 x 1.20	1	-	-	-	5.40
	Total quantity					5.40
22	Ventilator V (wooden frame with louvers) (m2)					

	1 x 1 No x 0.60 x 0.45	1	-	-	-	0.27
	Total quantity					0.27
23	Indian / European WC with flush tank (No)					
	Counted from plan, 1 No	1	-	-	-	1
	Total quantity					1
24	Wash basin with pedestal and tap (No)					
	Counted from plan, 1 No	1	-	-	-	1
	Total quantity					1
25	Kitchen sink (stainless steel, with drainboard) (No)					
	Counted from plan, 1 No	1	-	-	-	1
	Total quantity					1
26	CP tap / bib cock (15 mm) (No)					
	Counted from plan, 4 No	4	-	-	-	4
	Total quantity					4
27	Water supply and drainage pipes (lump sum, per floor) (LS)					
	Counted from plan, 1 LS	1	-	-	-	1
	Total quantity					1
28	Light point (concealed wiring) (point)					
	Counted from plan, 9 point	9	-	-	-	9
	Total quantity					9
29	Fan point (concealed wiring) (point)					
	Counted from plan, 3 point	3	-	-	-	3
	Total quantity					3
30	Power socket 5 A (with wiring) (point)					

	Counted from plan, 6 point	6	-	-	-	6
	Total quantity					6
31	Power socket 15 A (with wiring) (point)					
	Counted from plan, 1 point	1	-	-	-	1
	Total quantity					1
32	Distribution board with MCBs (No)					
	Counted from plan, 1 No	1	-	-	-	1
	Total quantity					1

Result

- The detailed estimate (quantities) of the single-storied building is prepared as above. Corrected centre line lengths: trench 35.50 m, footing 35.92 m and 36.38 m, wall 36.84 m.

Experiment 4 - Detailed Estimate of a Two-Storied Building by Centre Li

Aim

To prepare the detailed estimate of a two-storied (ground floor + first floor) residential building using the centre line method.

Requirements

- Same plan as the single storey building repeated on the first floor, with an RCC dog-legged staircase on the rear side, parapet wall 1.0 m high.

Theory

- The foundation is deeper and wider for the extra load: trench 1.10 m wide x 1.20 m deep, PCC 0.30 m, three footing steps (0.92, 0.69 and 0.46 m wide, 0.30 m each).
- Ground floor wall height = 3.15 m (plinth to slab bottom), first floor wall height = 3.00 m. Ground floor slab acts as the first floor slab. Plinth is 0.45 m above ground.

- Corrected centre line length L(w) = 37.30 - 2w. For w = 1.10, 0.92, 0.69, 0.46, 0.23 m the lengths are 35.10, 35.46, 35.92, 36.38 and 36.84 m.
- Quantities of the first floor (brickwork, plaster, paint, flooring, doors, windows, fittings) are taken in the same way as the ground floor.

Measurement sheet (two storey)

Sl	Particulars of item	No	L (m)	B (m)	D / H (m)	Quantity
1	Earthwork excavation in foundation (m3)					
	Earthwork (centre line)	1	35.10	1.10	1.20	46.33
	Total quantity					46.33
2	PCC 1:4:8 bed in foundation (m3)					
	PCC (centre line)	1	35.10	1.10	0.30	11.58
	Total quantity					11.58
3	Brickwork in CM 1:6, footing step 1 (width 0.92 m) (m3)					
	Footing 1 (centre line)	1	35.46	0.92	0.30	9.79
	Total quantity					9.79
4	Brickwork in CM 1:6, footing step 2 (width 0.69 m) (m3)					
	Footing 2 (centre line)	1	35.92	0.69	0.30	7.44
	Total quantity					7.44
5	Brickwork in CM 1:6, footing step 3 (width 0.46 m) (m3)					
	Footing 3 (centre line)	1	36.38	0.46	0.30	5.02
	Total quantity					5.02
6	Brickwork in CM 1:6 in basement up to plinth level (m3)					

	Basement (centre line)	1	36.84	0.23	0.45	3.81
	Total quantity					3.81
7	Earth filling in plinth, 0.35 m (m3)					
	Floor area (clear) 36.75 m2 x 0.35 m	1	-	36.75	0.35	12.86
	Total quantity					12.86
8	DPC 25 mm thick, CM 1:3 at plinth level (m2)					
	DPC (centre line)	1	36.84	0.23	-	8.47
	Total quantity					8.47
9	Brickwork in CM 1:6 in superstructure (m3)					
	Ground floor: gross masonry (centre line)	1	36.84	0.23	3.15	26.69
	Deduct D1 (external door)	1	1.00	0.23	2.10	-0.48
	Deduct D2 (internal door)	2	0.90	0.23	2.10	-0.87
	Deduct W (window)	3	1.50	0.23	1.20	-1.24
	Deduct V (ventilator)	1	0.60	0.23	0.45	-0.06
	Deduct RCC lintels	1	10.00	0.23	0.15	-0.35
	First floor: gross masonry (centre line)	1	36.84	0.23	3.00	25.42
	Deduct D1 (external door)	1	1.00	0.23	2.10	-0.48
	Deduct D2 (internal door)	2	0.90	0.23	2.10	-0.87
	Deduct W (window)	3	1.50	0.23	1.20	-1.24
	Deduct V (ventilator)	1	0.60	0.23	0.45	-0.06
	Deduct RCC lintels	1	10.00	0.23	0.15	-0.35
	Total quantity					46.11
10	Brickwork in CM 1:6 in parapet wall, 1.0 m high (m3)					
	Parapet (centre line perimeter 29.84)	1	29.84	0.23	1.00	6.86

	Total quantity					6.86
11	RCC 1:2:4 in lintels (m3)					
	Lintels, 2 floor(s)	2	10.00	0.23	0.15	0.69
	Total quantity					0.69
12	RCC 1:2:4 in roof / floor slab, 120 mm thick (m3)					
	Slab 11.42 x 3.96 x 0.12, 2 Nos	2	11.42	3.96	0.12	10.85
	Total quantity					10.85
13	RCC 1:2:4 in staircase (dog-legged) (m3)					
	Waist slab, 2 flights x 3.40 x 1.00 x 0.15	2	3.40	1.00	0.15	1.02
	Steps, 20 Nos x (1/2 x 0.25 x 0.175 x 1.00)	20	0.25	0.175	1.00	0.44
	Mid-landing 1.20 x 1.00 x 0.15	1	1.20	1.00	0.15	0.18
	Total quantity					1.64
14	Reinforcement steel Fe 500 @ 100 kg per m3 of RCC (kg)					
	RCC 13.181 m3 x 100 kg	1	-	-	-	1318.1
	Total quantity					1318.11
15	Centering and shuttering (m2)					
	Slab soffit 36.75 m2, 2 Nos	2	-	36.75	-	73.50
	Slab edges 30.76 x 0.12, 2 Nos	2	30.76	0.12	-	7.38
	Lintels 10.0 x (0.23 + 2 x 0.15), 2 floor(s)	2	10.00	0.53	-	10.60
	Staircase (waist soffit 6.8 + risers 3.5 + landing 1.2)	1	-	-	-	11.50
	Total quantity					102.98
16	Cement concrete flooring 40 mm thick (1:2:4) (m2)					
	Floor area, 2 floor(s)	2	-	36.75	-	73.50

	Total quantity					73.50
17	Plastering 12 mm thick, CM 1:6, internal walls (m2)					
	GF Hall (4.0 x 3.5), perimeter 15.00	1	15.00	-	3.15	47.25
	GF Bedroom (3.5 x 3.5), perimeter 14.00	1	14.00	-	3.15	44.10
	GF Kitchen (3.0 x 3.5), perimeter 13.00	1	13.00	-	3.15	40.95
	Deduct D1 (external door), 1 No x 1 face(s)	1	1.00	-	2.10	-2.10
	Deduct D2 (internal door), 2 No x 2 face(s)	4	0.90	-	2.10	-7.56
	Deduct W (window), 3 No x 1 face(s)	3	1.50	-	1.20	-5.40
	Deduct V (ventilator), 1 No x 1 face(s)	1	0.60	-	0.45	-0.27
	FF Hall (4.0 x 3.5), perimeter 15.00	1	15.00	-	3.00	45.00
	FF Bedroom (3.5 x 3.5), perimeter 14.00	1	14.00	-	3.00	42.00
	FF Kitchen (3.0 x 3.5), perimeter 13.00	1	13.00	-	3.00	39.00
	Deduct D1 (external door), 1 No x 1 face(s)	1	1.00	-	2.10	-2.10
	Deduct D2 (internal door), 2 No x 2 face(s)	4	0.90	-	2.10	-7.56
	Deduct W (window), 3 No x 1 face(s)	3	1.50	-	1.20	-5.40
	Deduct V (ventilator), 1 No x 1 face(s)	1	0.60	-	0.45	-0.27
	Total quantity					227.64
18	Plastering 12 mm thick, CM 1:6, ceilings (m2)					
	Ceiling area, 2 floor(s)	2	-	36.75	-	73.50
	Total quantity					73.50
19	Plastering 15 mm thick, CM 1:6, external (m2)					
	External face, perimeter 30.76 x 7.72	1	30.76	-	7.72	237.47

	Deduct openings (D1, D2 x1 face, W, V) x 2	2	-	-	-	-23.10
	Parapet inner face 29.84 x 1.00	1	29.84	-	1.00	29.84
	Total quantity					244.21
20	Emulsion paint, 2 coats over primer (walls and ceilings) (m2)					
	Internal plaster + ceiling plaster areas	1	-	-	-	301.14
	Total quantity					301.14
21	Weather-proof exterior paint, 2 coats (m2)					
	External plaster area	1	-	-	-	244.21
	Total quantity					244.21
22	Door D1 (panelled, teak, with frame and fixtures) (m2)					
	2 x 1 No x 1.00 x 2.10	1	-	-	-	4.20
	Total quantity					4.20
23	Door D2 (flush, with frame and fixtures) (m2)					
	2 x 2 No x 0.90 x 2.10	1	-	-	-	7.56
	Total quantity					7.56
24	Window W (wooden, glazed with grill) (m2)					
	2 x 3 No x 1.50 x 1.20	1	-	-	-	10.80
	Total quantity					10.80
25	Ventilator V (wooden frame with louvers) (m2)					
	2 x 1 No x 0.60 x 0.45	1	-	-	-	0.54
	Total quantity					0.54

26	Indian / European WC with flush tank (No)					
	Counted from plan, 2 No	2	-	-	-	2
	Total quantity					2
27	Wash basin with pedestal and tap (No)					
	Counted from plan, 2 No	2	-	-	-	2
	Total quantity					2
28	Kitchen sink (stainless steel, with drainboard) (No)					
	Counted from plan, 1 No	1	-	-	-	1
	Total quantity					1
29	CP tap / bib cock (15 mm) (No)					
	Counted from plan, 7 No	7	-	-	-	7
	Total quantity					7
30	Water supply and drainage pipes (lump sum, per floor) (LS)					
	Counted from plan, 2 LS	2	-	-	-	2
	Total quantity					2
31	Light point (concealed wiring) (point)					
	Counted from plan, 18 point	18	-	-	-	18
	Total quantity					18
32	Fan point (concealed wiring) (point)					
	Counted from plan, 6 point	6	-	-	-	6
	Total quantity					6
33	Power socket 5 A (with wiring) (point)					
	Counted from plan, 12 point	12	-	-	-	12
	Total quantity					12

34	Power socket 15 A (with wiring) (point)					
	Counted from plan, 2 point	2	-	-	-	2
	Total quantity					2
35	Distribution board with MCBs (No)					
	Counted from plan, 2 No	2	-	-	-	2
	Total quantity					2

Result

- Detailed estimate of the two-storied building is prepared. Brickwork in superstructure = 46.11 m3 and RCC = 13.18 m3.

Experiment 5 – Activity Based Learning of Estimate Data and Rate Analy

Aim

To study the data for estimating (quantities of materials and labour per unit of work) and to prepare the rate analysis of the main items of building work.

Requirements

- Standard Data Book / SSR, current prices of materials and wages, calculator.

Theory

- Rate analysis is the process of finding the cost of one unit of an item of work by adding the cost of materials, labour, machinery and the contractor's profit and overheads.
- Analysis of rate = Materials + Labour + Machinery / tools + Water charges and scaffolding + Contractor's profit and overheads (taken as 10 % here).
- **Sources of data** Standard Data Book, SSR, IS 1200 and market prices; lead and lift charges are added where required.

Assumed basic rates Cement Rs. 410 per bag (50 kg); Sand Rs. 1,900 per m3; 40 mm metal Rs. 1,650 per m3; 20 mm metal Rs. 1,750 per m3; 12 mm metal Rs. 1,800 per m3; First class brick Rs. 9 each; Steel Fe 500 Rs. 72 per kg; Binding wire Rs. 90 per kg. Wages per day :: Mason Rs. 900, Mazdoor Rs. 600, Carpenter Rs. 950, Bar bender Rs. 900, Painter Rs. 850.

Procedure

- Select the item and the unit of measurement (as per IS 1200).
- List the materials and labour required per unit from the data book.
- Multiply by the current rates, add them, add 10 % for profit and overheads and round off.

5.1 Earthwork excavation (ordinary soil, up to 1.5 m) - per m3

Sl	Particulars	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	Mazdoor	0.45	day	600.00	270.00
	Cost				270.00
	Add 10 % contractor's profit and overheads				27.00
	RATE per m3 (rounded)				297.00

5.2 PCC 1:4:8 (40 mm metal) - per m3

Sl	Particulars	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	Cement	3.4	bag	410.00	1,394.00
2	Sand	0.48	m3	1,900.00	912.00
3	40 mm metal	0.95	m3	1,650.00	1,567.50
4	Mason	0.4	day	900.00	360.00
5	Mazdoor	3.0	day	600.00	1,800.00
	Cost				6,033.50
	Add 10 % contractor's profit and overheads				603.35

	RATE per m3 (rounded)				6,637.00

5.3 Brickwork in CM 1:6 (foundation and basement) - per m3

Sl	Particulars	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	First class bricks	500	No	9.00	4,500.00
2	Cement (mortar 0.25 m3)	1.3	bag	410.00	533.00
3	Sand	0.27	m3	1,900.00	513.00
4	Mason	1.0	day	900.00	900.00
5	Mazdoor	1.8	day	600.00	1,080.00
	Cost				7,526.00
	Add 10 % contractor's profit and overheads				752.60
	RATE per m3 (rounded)				8,279.00

5.4 Brickwork in CM 1:6 (superstructure) - per m3

Sl	Particulars	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	First class bricks	500	No	9.00	4,500.00
2	Cement (mortar 0.25 m3)	1.3	bag	410.00	533.00
3	Sand	0.27	m3	1,900.00	513.00
4	Mason	1.0	day	900.00	900.00
5	Mazdoor	1.8	day	600.00	1,080.00
6	Scaffolding (lump)	1	LS	150.00	150.00
	Cost				7,676.00
	Add 10 % contractor's profit and overheads				767.60
	RATE per m3 (rounded)				8,444.00

5.5 RCC 1:2:4 (20 mm metal) excluding steel and shuttering - per m3

Sl	Particulars	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	Cement	6.3	bag	410.00	2,583.00
2	Sand	0.44	m3	1,900.00	836.00
3	20 mm metal	0.88	m3	1,750.00	1,540.00
4	Mason	1.0	day	900.00	900.00
5	Mazdoor	4.0	day	600.00	2,400.00
6	Mixer and vibrator (lump)	1	LS	300.00	300.00
	Cost				8,559.00
	Add 10 % contractor's profit and overheads				855.90
	RATE per m3 (rounded)				9,415.00

5.6 Reinforcement Fe 500 (cutting, bending, placing) - per kg

Sl	Particulars	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	Steel Fe 500	1.0	kg	72.00	72.00
2	Binding wire	0.01	kg	90.00	0.90
3	Bar bender	0.012	day	900.00	10.80
4	Mazdoor	0.01	day	600.00	6.00
	Cost				89.70
	Add 10 % contractor's profit and overheads				8.97
	RATE per kg (rounded)				99.00

5.7 Centering and shuttering - per m2

Sl	Particulars	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	Shuttering material (lump)	1	LS	95.00	95.00
2	Carpenter	0.2	day	950.00	190.00
3	Mazdoor	0.2	day	600.00	120.00

	Cost				405.00
	Add 10 % contractor's profit and overheads				40.50
	RATE per m2 (rounded)				446.00

5.8 Earth filling in plinth with compaction - per m3

Sl	Particulars	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	Mazdoor	0.6	day	600.00	360.00
2	Watering and compaction (lump)	1	LS	40.00	40.00
	Cost				400.00
	Add 10 % contractor's profit and overheads				40.00
	RATE per m3 (rounded)				440.00

5.9 DPC 25 mm thick, CM 1:3 with waterproofing compound - per m2

Sl	Particulars	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	Cement	0.25	bag	410.00	102.50
2	Sand	0.026	m3	1,900.00	49.40
3	Waterproofing compound	1	LS	25.00	25.00
4	Mason	0.1	day	900.00	90.00
5	Mazdoor	0.1	day	600.00	60.00
	Cost				326.90
	Add 10 % contractor's profit and overheads				32.69
	RATE per m2 (rounded)				360.00

5.10 Plaster 12 mm thick, CM 1:6 (internal and ceiling) - per m2

Sl	Particulars	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	Cement	0.078	bag	410.00	31.98
2	Sand	0.0164	m3	1,900.00	31.16

3	Mason	0.14	day	900.00	126.00
4	Mazdoor	0.14	day	600.00	84.00
5	Scaffolding (lump)	1	LS	6.00	6.00
	Cost				279.14
	Add 10 % contractor's profit and overheads				27.91
	RATE per m2 (rounded)				307.00

5.11 Plaster 15 mm thick, CM 1:6 (external) - per m2

Sl	Particulars	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	Cement	0.0988	bag	410.00	40.51
2	Sand	0.0207	m3	1,900.00	39.33
3	Mason	0.16	day	900.00	144.00
4	Mazdoor	0.16	day	600.00	96.00
5	Scaffolding (lump)	1	LS	10.00	10.00
	Cost				329.84
	Add 10 % contractor's profit and overheads				32.98
	RATE per m2 (rounded)				363.00

5.12 Cement concrete flooring 40 mm thick, 1:2:4 - per m2

Sl	Particulars	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	Cement (concrete and slurry)	0.3	bag	410.00	123.00
2	Sand	0.0194	m3	1,900.00	36.86
3	12 mm metal	0.0387	m3	1,800.00	69.66
4	Mason	0.12	day	900.00	108.00
5	Mazdoor	0.2	day	600.00	120.00
	Cost				457.52
	Add 10 % contractor's profit and overheads				45.75

	RATE per m2 (rounded)				503.00

S.13 Emulsion paint, 2 coats over primer (internal) - per m2

Sl	Particulars	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	Primer and emulsion paint	1	LS	150.00	150.00
2	Painter	0.1	day	850.00	85.00
3	Mazdoor	0.05	day	600.00	30.00
	Cost				265.00
	Add 10 % contractor's profit and overheads				26.50
	RATE per m2 (rounded)				292.00

S.14 Weather-proof exterior paint, 2 coats (external) - per m2

Sl	Particulars	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	Primer and exterior paint	1	LS	190.00	190.00
2	Painter	0.12	day	850.00	102.00
3	Mazdoor	0.06	day	600.00	36.00
4	Scaffolding (lump)	1	LS	10.00	10.00
	Cost				338.00
	Add 10 % contractor's profit and overheads				33.80
	RATE per m2 (rounded)				372.00

Result

- Rate analysis of the 14 main items of building work was done as above; the rates are used in the abstract estimates of Exercises 6 and 7. Ready-made items (doors, windows, sanitary and electrical fittings) are taken from the market rates / SSR.

Experiment 6 – Abstract Estimate for the Detailed Estimate in Exercise

Aim

To prepare the abstract estimate (with rates and amounts) for the single-storied building of Exercise 3.

Theory

- The abstract estimate is prepared by multiplying the total quantity of each item (from the detailed estimate) with its rate (from the rate analysis of Exercise 5) and adding all the amounts.
- Contingencies (3 %) are added for unforeseen items, and work-charged establishment (1.5 %) for supervision and watchmen.
- Sl. No. of the items follows the order of the detailed estimate of Exercise 3.

Abstract of estimate

Sl	Description of item	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	Earthwork excavation in foundation	31.95	m3	297.00	9,489.15
2	PCC 1:4:8 bed in foundation	9.58	m3	6,637.00	63,615.64
3	Brickwork in CM 1:6, footing step 1 (width 0.69 m)	7.44	m3	8,279.00	61,558.01
4	Brickwork in CM 1:6, footing step 2 (width 0.46 m)	5.02	m3	8,279.00	41,564.22
5	Brickwork in CM 1:6 in basement up to plinth level	4.66	m3	8,279.00	38,582.29
6	Earth filling in plinth, 0.35 m	12.86	m3	440.00	5,659.50
7	DPC 25 mm thick, CM 1:3 at plinth level	8.47	m2	360.00	3,050.35
8	Brickwork in CM 1:6 in superstructure	23.69	m3	8,444.00	2,00,030.59
9	RCC 1:2:4 in lintels	0.35	m3	9,415.00	3,248.18
10	RCC 1:2:4 in roof / floor slab, 120 mm thick	5.43	m3	9,415.00	51,093.17

11	Reinforcement steel Fe 500 @ 100 kg per m3 of RCC	577.18	kg	99.00	57,140.66
12	Centering and shuttering	45.74	m2	446.00	20,400.04
13	Cement concrete flooring 40 mm thick (1:2:4)	36.75	m2	503.00	18,485.25
14	Plastering 12 mm thick, CM 1:6, internal walls	116.97	m2	307.00	35,909.79
15	Plastering 12 mm thick, CM 1:6, ceilings	36.75	m2	307.00	11,282.25
16	Plastering 15 mm thick, CM 1:6, external	99.19	m2	363.00	36,004.52
17	Emulsion paint, 2 coats over primer (walls and ceilings)	153.72	m2	292.00	44,886.24
18	Weather-proof exterior paint, 2 coats	99.19	m2	372.00	36,897.19
19	Door D1 (panelled, teak, with frame and fixtures)	2.10	m2	14,000.00	29,400.00
20	Door D2 (flush, with frame and fixtures)	3.78	m2	7,800.00	29,484.00
21	Window W (wooden, glazed with grill)	5.40	m2	8,800.00	47,520.00
22	Ventilator V (wooden frame with louvers)	0.27	m2	7,200.00	1,944.00
23	Indian / European WC with flush tank	1	No	7,100.00	7,100.00
24	Wash basin with pedestal and tap	1	No	3,800.00	3,800.00
25	Kitchen sink (stainless steel, with drainboard)	1	No	5,200.00	5,200.00
26	CP tap / bib cock (15 mm)	4	No	450.00	1,800.00
27	Water supply and drainage pipes (lump sum, per floor)	1	LS	22,000.00	22,000.00
28	Light point (concealed wiring)	9	point	1,250.00	11,250.00
29	Fan point (concealed wiring)	3	point	1,350.00	4,050.00
30	Power socket 5 A (with wiring)	6	point	880.00	5,280.00

31	Power socket 15 A (with wiring)	1	point	1,250.00	1,250.00
32	Distribution board with MCBs	1	No	6,500.00	6,500.00
	Total of items				9,15,475.05
	Add contingencies @ 3 %				27,464.25
	Add work-charged establishment @ 1.5 %				13,732.13
	GRAND TOTAL (Estimate amount)				9,56,671.43

Result

- The estimated cost of the single-storied building = Rs. 9,56,671.43 (Rupees 9,56,671 only).
- Cost per m2 of plinth area = Rs. 21,154 (approx.).

Experiment 7 – Abstract Estimate for the Detailed Estimate in Exercise 4

Aim

To prepare the abstract estimate for the two-storied building of Exercise 4.

Theory

- The abstract estimate is prepared by multiplying the total quantity of each item (from the detailed estimate) with its rate (from the rate analysis of Exercise 5) and adding all the amounts.
- Contingencies (3 %) are added for unforeseen items, and work-charged establishment (1.5 %) for supervision and watchmen.
- Sl. No. of the items follows the order of the detailed estimate of Exercise 4.

Abstract of estimate

Sl	Description of item	Quantity	Unit	Rate (Rs.)	Amount (Rs.)
1	Earthwork excavation in foundation	46.33	m3	297.00	13,760.60
2	PCC 1:4:8 bed in foundation	11.58	m3	6,637.00	76,876.37

3	Brickwork in CM 1:6, footing step 1 (width 0.92 m)	9.79	m3	8,279.00	81,026.24
4	Brickwork in CM 1:6, footing step 2 (width 0.69 m)	7.44	m3	8,279.00	61,558.01
5	Brickwork in CM 1:6, footing step 3 (width 0.46 m)	5.02	m3	8,279.00	41,564.22
6	Brickwork in CM 1:6 in basement up to plinth level	3.81	m3	8,279.00	31,567.33
7	Earth filling in plinth, 0.35 m	12.86	m3	440.00	5,659.50
8	DPC 25 mm thick, CM 1:3 at plinth level	8.47	m2	360.00	3,050.35
9	Brickwork in CM 1:6 in superstructure	46.11	m3	8,444.00	3,89,329.03
10	Brickwork in CM 1:6 in parapet wall, 1.0 m high	6.86	m3	8,444.00	57,952.86
11	RCC 1:2:4 in lintels	0.69	m3	9,415.00	6,496.35
12	RCC 1:2:4 in roof / floor slab, 120 mm thick	10.85	m3	9,415.00	1,02,186.34
13	RCC 1:2:4 in staircase (dog-legged)	1.64	m3	9,415.00	15,417.06
14	Reinforcement steel Fe 500 @ 100 kg per m3 of RCC	1318.11	kg	99.00	1,30,492.57
15	Centering and shuttering	102.98	m2	446.00	45,929.08
16	Cement concrete flooring 40 mm thick (1:2:4)	73.50	m2	503.00	36,970.50
17	Plastering 12 mm thick, CM 1:6, internal walls	227.64	m2	307.00	69,885.48
18	Plastering 12 mm thick, CM 1:6, ceilings	73.50	m2	307.00	22,564.50
19	Plastering 15 mm thick, CM 1:6, external	244.21	m2	363.00	88,647.21
20	Emulsion paint, 2 coats over primer (walls and ceilings)	301.14	m2	292.00	87,932.88

21	Weather-proof exterior paint, 2 coats	244.21	m2	372.00	90,845.08
22	Door D1 (panelled, teak, with frame and fixtures)	4.20	m2	14,000.00	58,800.00
23	Door D2 (flush, with frame and fixtures)	7.56	m2	7,800.00	58,968.00
24	Window W (wooden, glazed with grill)	10.80	m2	8,800.00	95,040.00
25	Ventilator V (wooden frame with louvers)	0.54	m2	7,200.00	3,888.00
26	Indian / European WC with flush tank	2	No	7,100.00	14,200.00
27	Wash basin with pedestal and tap	2	No	3,800.00	7,600.00
28	Kitchen sink (stainless steel, with drainboard)	1	No	5,200.00	5,200.00
29	CP tap / bib cock (15 mm)	7	No	450.00	3,150.00
30	Water supply and drainage pipes (lump sum, per floor)	2	LS	22,000.00	44,000.00
31	Light point (concealed wiring)	18	point	1,250.00	22,500.00
32	Fan point (concealed wiring)	6	point	1,350.00	8,100.00
33	Power socket 5 A (with wiring)	12	point	880.00	10,560.00
34	Power socket 15 A (with wiring)	2	point	1,250.00	2,500.00
35	Distribution board with MCBs	2	No	6,500.00	13,000.00
	Total of items				18,07,217.58
	Add contingencies @ 3 %				54,216.53
	Add work-charged establishment @ 1.5 %				27,108.26
	GRAND TOTAL (Estimate amount)				18,88,542.37

Result

- The estimated cost of the two-storied building = Rs. 18,88,542.37 (Rupees 18,88,542 only).
- Cost per m2 of plinth area = Rs. 20,880 (approx.).

Experiment 8 – Measurement Book and Bill Preparation (AP State Governm

Aim

To write the Measurement Book (MB) entries for the work of Exercise 3 and to prepare the contractor's bill from the abstract estimate of Exercise 6.

Requirements

- Measurement Book (MB), agreement / work order, sanctioned estimate, bill (Form), calculator.

Theory

- **Measurement Book (MB)** the basic record of all the measurements of works actually done. Each page is numbered, entries are made in ink, no page is torn out and corrections are made by cutting and initialling. Measurements are recorded by the Assistant Engineer / Works Inspector, then checked by the superior officers as per the check measurement percentages prescribed by the department.
- **Each entry gives** date of measurement, description of the item as in the agreement, the dimensions (No x L x B x D), quantity, and the signature of the recording officer and the contractor's representative.
- **Bill** prepared from the MB measurements and the agreed rates. Types are Running Account (RA) bills (part payments during work) and the Final bill (last payment). The abstract of the bill is passed by the Engineer and paid by the Treasury / Pay and Accounts office.
- **Deductions in the bill (as applicable)** security deposit, income tax at source, labour cess, GST at source, recovery of advances and materials supplied by the department.

Particulars of the work

- **Name of work** Construction of a single-storied residential building (as per Exercises 3 and 6). Agreement amount = Rs. 9,56,671.43. Name of contractor

:: Sri / M/s ____ (sample). MB No. 214, Pages 31 to 36 (sample).

MB entries (sample)

Date	Description of work and measurement	No	L	B	D	Quantity
12-01-2026	Earthwork excavation (Item 1)	1	35.50	0.90	1.00	31.95
16-01-2026	PCC 1:4:8 (Item 2)	1	35.50	0.90	0.30	9.58
28-01-2026	Brickwork in basement (Item 5)	1	36.84	0.23	0.55	4.66
25-02-2026	Superstructure brickwork (Item 8), ground floor	1	36.84	0.23	3.15	23.69

Running Account Bill No. 1 (sample)

Sl	Description of item	Quantity executed up to date	Unit	Rate (Rs.)	Amount (Rs.)
1	Earthwork excavation in foundation	31.95	m3	297.00	9,489.15
2	PCC 1:4:8 bed in foundation	9.58	m3	6,637.00	63,615.64
3	Brickwork in CM 1:6, footing step 1 (width 0.69 m)	7.44	m3	8,279.00	61,558.01
4	Brickwork in CM 1:6, footing step 2 (width 0.46 m)	5.02	m3	8,279.00	41,564.22
5	Brickwork in CM 1:6 in basement up to plinth level	4.66	m3	8,279.00	38,582.29
6	Earth filling in plinth, 0.35 m	12.86	m3	440.00	5,659.50
7	DPC 25 mm thick, CM 1:3 at plinth level	8.47	m2	360.00	3,050.35
8	Brickwork in CM 1:6 in superstructure (60 % of work done)	14.21	m3	8,444.00	1,20,018.35
	Gross amount of work done up to date				3,43,537.52

	Less: previous payments (this is the Ist running bill)				0.00
	Amount of this bill				3,43,537.52
	Less: Security deposit @ 5 %				17,176.88
	Less: Income tax (TDS) @ 2 %				6,870.75
	Less: Labour welfare cess @ 1 %				3,435.38
	Less: GST (TDS) @ 2 %				6,870.75
	NET AMOUNT PAYABLE				3,09,183.77

Result

- The measurement book entries and the first running account bill were prepared. Net amount payable to the contractor = Rs. 3,09,183.77.
- **Precautions** record measurements on the same day, never leave blank spaces, get the contractor's signature, and follow the sequence of items in the agreement.

Experiment 9 – Detailed Specifications and Model Tender Document

Aim

To write the detailed specifications for the main items of Exercises 3 and 6 and to prepare a model tender document for the work.

Requirements

- IS codes (IS 456, IS 1077, IS 269, IS 383, IS 2212, IS 1200), PWD specifications, SSR, tender format.

Theory

Specifications describe the quality of materials, the proportions, the method of work, the tests and the method of measurement and payment. They are of two types: general specifications (the nature of the whole work) and detailed specifications (each item).

Detailed specifications of main items

Item 1 – Earthwork excavation in foundation

- Excavate in ordinary soil to the lines, levels and dimensions shown on the drawing; the sides are kept vertical or timbered where required. Remove roots and rubbish; dress and level the bottom and ram it.
- Excavated earth is stacked at 1.5 m away from the edge; surplus soil is removed to the place directed up to 50 m lead and 1.5 m lift.
- **Measurement** in cubic metre of the net volume (L x B x D). Payment at the rate for the item.

Item 2 – Plain cement concrete 1:4:8

- Cement OPC 43 grade (IS 269), coarse sand (zone II, IS 383) and 40 mm graded hard broken stone metal. Proportion by volume 1 : 4 : 8; water cement ratio not more than 0.7.
- Mixing in a mechanical mixer for at least 2 minutes; concrete placed in 150 mm layers and compacted; cured for 7 days. Thickness as per drawing.
- **Measurement** cubic metre; deductions are not made for openings less than 0.1 m3.

Item 3 – Brickwork in CM 1:6 (foundation, basement, superstructure)

- Bricks shall be first class, well burnt, of size 230 x 110 x 70 mm (nominal 190 x 90 x 90), crushing strength not less than 35 kg/cm2, water absorption not more than 20 % (IS 1077). Bricks soaked in water for at least 2 hours before use.
- Mortar 1 cement : 6 sand by volume mixed dry and then with water; used within 30 minutes. Joint thickness 10 mm; English bond; courses laid level and plumb; vertical joints broken; raking of joints 15 mm deep for plastering.
- Cured for at least 10 days. Work is raised in uniform layers, not more than 1.0 m in a day.

- **Measurement** cubic metre; openings deducted as per IS 1200 (deduct all openings above 0.1 m2; lintel bearings as per drawings).

Item 4 – RCC 1:2:4 (lintel, slab, staircase)

- Cement OPC 43 grade, sand and 20 mm graded aggregates; grade M20 (IS 456); water cement ratio 0.5; slump 50 to 75 mm.
- Reinforcement Fe 500 conforming to IS 1786, free from rust and oil; cover 20 mm for slab and 25 mm for beams and lintels, with cover blocks; laps as per IS 456.
- Shuttering of steel or plywood, rigid and water tight; stripping of forms after 7 days for slab sides and 14 days for soffit with props (21 days for spans above 4.5 m). Concrete cured for 14 days.
- **Measurement** concrete in cubic metre (no deduction for steel); steel in kg from the bar bending schedule; shuttering in square metre of contact surface.

Item 5 – Plastering in CM 1:6

- Surface cleaned, joints raked and wetted. Mortar 1:6 applied in two coats to a thickness of 12 mm (15 mm for external); the surface is finished plumb and true, cured for 7 days.
- **Measurement** square metre; openings up to 0.5 m2 not deducted.

Item 6 – Cement concrete flooring (40 mm)

- Sub base of well compacted earth and 100 mm lime or cement concrete; 40 mm thick concrete 1:2:4 with 12 mm stone chips, laid in panels of 2 m x 2 m with glass or aluminium strips, finished with neat cement punning and cured for 10 days.
- **Measurement** square metre of finished floor.

Item 7 – Doors, windows, painting, sanitary and electrical works

- Wood seasoned (teak or country wood) as per IS 1003; frame and shutters as per drawing with iron fittings (hinges, tower bolts, aldrops) and one coat of

primer; measured in square metre of shutter and frame area.

- Emulsion paint in two coats over one coat of primer on a smooth surface; exterior weather-proof paint in two coats. Measurement: square metre.
- Sanitary fittings of ISI mark (IS 2556 for WC and wash basin), fixed on proper bed with CP fittings; measured in numbers.
- Electrical wiring in concealed PVC conduit with copper wires of 1.5 sq mm and 2.5 sq mm (IS 694) with ISI mark switches and MCBs; measured in points.

Model Tender Document (extract)

Particular	Details (sample)
Name of work	Construction of a residential building (G) as per drawings
Estimated contract value (put to tender)	Rs. 9,56,671.43
Earnest Money Deposit (EMD)	1 % of estimated value (sample) = Rs. 9,566.71
Period of completion	6 months from the date of agreement
Defect liability period	12 months after completion
Security deposit	5 % of the bill amount (recovered from the running bills)
Performance security	As per department norms
Mode of tender	e-Procurement (online bids)
Validity of tender	90 days from the date of opening
Eligibility	Registered contractor of suitable class with experience of similar works
Last date of submission	30 days from the date of Notice Inviting Tender

Contents of the tender document

- **Part A** Notice Inviting Tender (NIT), Instructions to bidders, qualification requirements, form of tender and bid.
- **Part B** Conditions of contract (time, extension, penalty or liquidated damages,

payment, deductions, termination, arbitration, defect liability, price variation clause if allowed).

- **Part C** Technical specifications (as above) and drawings.
- **Part D** Bill of Quantities / Schedule of items with quantities (from Exercise 6) in which the bidder quotes the rates (percentage above or below the estimate).

Result

- The detailed specifications were written for the main items and the model tender document was prepared for the work estimated at Rs. 9,56,671.43.

Experiment 10 – Valuation of Buildings, Cost Escalation Procedures and Va

Aim

To carry out the valuation of the single-storied building (Exercise 6), to calculate the cost escalation of the work and to do the value analysis of the walling material.

Requirements

- Estimate / completion cost, age and life of the building, rent data, price indices, calculator.

Theory

- **Valuation** the technique of estimating the fair price or value of a property. Cost = original cost; Value = worth of the property at a given date. Types :: market value, book value, scrap value, salvage value, rental value, distress value.
- **Methods of valuation** (1) Rental return method (capitalised value of net income = net annual income x years purchase), (2) Cost (depreciation) method :: value = cost of building - depreciation + value of land, (3) Development method, (4) Belting method, (5) Profit based method.
- **Depreciation** loss of value due to age, wear and obsolescence. Methods ::

Straight line, Declining balance (constant percentage), Sinking fund and Quantity survey method.

Formulas Straight line :: annual depreciation D = (C - S)/n, value after t years = C - D t. Declining balance :: rate r = 1 - (S/C) to the power 1/n, value = C (1 - r) to the power t. Sinking fund :: annual instalment = (C - S) x i / [(1 + i)^n - 1]. Years Purchase (YP) = 100 / rate of return.

Procedure and data

- Building cost (from Exercise 6) C = Rs. 9,56,671.43; life of RCC building n = 60 years; age t = 12 years; scrap value S = 10 % of C = Rs. 95,667.14.
- **Land** 150 m2 at Rs. 14,000 per m2 = Rs. 21,00,000.00. Monthly rent of the building = Rs. 16,000; outgoings (tax, repairs, insurance, vacancy) = 30 % of rent; rate of return = 7 %.

Valuation by cost (depreciation) method

Method	Calculation	Depreciated value of building (Rs.)
Straight line	D = (9,56,671.43 - 95,667.14) / 60 = 14,350.07 per year; value = C - D x 12	7,84,470.57
Declining balance	r = 1 - (0.10) to the power 1/60 = 3.76 % ; value = C (1 - r) to the power 12	6,03,618.86
Sinking fund (i = 5 %)	Annual instalment = 2,435.08 ; value = C - accumulated fund	9,17,911.97

- Total value (straight line) = value of building + land = 7,84,470.57 + 21,00,000.00 = Rs. 28,84,470.57.

Valuation by rental return method

Item	Amount (Rs.)
Gross annual rent (16,000 x 12)	1,92,000.00
Less outgoings @ 30 %	57,600.00

Net annual income	1,34,400.00
Years purchase at 7 % (= 100 / 7)	14.29
Capitalised value (net income x YP)	19,20,000.00

- Capital value of the property (building and land) by this method = Rs. 19,20,000.00.

Cost escalation

- Escalation is the increase in the cost of the work during the contract period because of the rise in price of labour, materials and fuel. It is allowed by the price adjustment (price variation) clause: Escalation = V x [sum of (weight x (It/I0 - 1))], where V = value of the work done in the period.
- Contract value V = Rs. 9,56,671.43; period = 12 months; indices (base and current) are sample values.

Component	Weight	Base index I0	Current index It	Increase (It/I0 - 1)	Weight x increase
Labour	0.3	100	108	0.0800	0.0240
Cement	0.15	100	112	0.1200	0.0180
Steel	0.15	100	105	0.0500	0.0075
Other materials	0.25	100	104	0.0400	0.0100
Fixed portion (no escalation)	0.15	-	-	-	0.0000

- Total price variation factor = 0.0595. Escalation amount = 9,56,671.43 x 0.0595 = Rs. 56,921.95 (5.95 % of contract value).
- Revised contract value = Rs. 10,13,593.38.

Value analysis (value engineering) of external wall

- Value analysis studies the function of an item, its cost and its worth, to get the required function at the lowest cost. Steps: information, function analysis, creativity (alternatives), evaluation and recommendation.

- **Weights** strength 0.35, thermal comfort 0.25, durability 0.20, speed of construction 0.20 (marks out of 10; the costs are assumed except for A).

Alternative	Cost (Rs. per m3)	Strength (w 0.35)	Thermal (w 0.25)	Durability (w 0.20)	Speed (w 0.20)	Weighte score	Value index = score x 1000 / cost
A First class brick masonry	8,444	8	6	8	5	6.90	0.817
B Fly ash brick masonry	7,650	8	7	8	7	7.55	0.987
C Solid concrete block	8,100	9	6	8	7	7.65	0.944
D AAC block masonry	6,900	6	9	7	9	7.55	1.094

Result

- Depreciated value of building (straight line) = Rs. 7,84,470.57, total value with land = Rs. 28,84,470.57; value by the rental return method = Rs. 19,20,000.00.
- Cost escalation = Rs. 56,921.95 (5.95 %).
- **Value analysis** alternative "AAC block masonry" gives the highest value index (1.094), so it is the best value for money among the four.

Summary of Important Formulas and Points

Topic	Key point
Centre line length	L(w) = sum of centre lengths - (number of junctions x w/2)
Long wall	L(c/c) + w ; Short wall = B(c/c) - w
Volume	No x L x B x D
Plinth area estimate	Plinth area x plinth area rate

Rate analysis	Materials + Labour + Machinery + 10 % profit and overheads
Abstract estimate	Quantity x rate for each item + contingencies + WCE
Depreciation (straight line)	D = (C - S) / n
Years purchase	YP = 100 / rate of return
Price escalation	V x sum of (weight x (It / 10 - 1))

Viva Voce Questions

- What is an estimate? Name the types of estimates.
- Differentiate between the wall to wall and the centre line methods.
- Why do we deduct half the wall thickness at each junction in the centre line method?
- What is a measurement book? Who records and who checks the measurements?
- What is rate analysis? What are the components of the rate of an item?
- What is the difference between a detailed estimate and an abstract estimate?
- Name the contents of a tender document. What is EMD?
- What is depreciation? Explain the straight line and sinking fund methods.
- What is cost escalation and how is it calculated?
- What is value analysis?